

TEJAS MANDWADE

Python Developer | Backend Developer | AI Engineer

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Professional Summary

Python-focused Software Engineer with hands-on experience building backend systems and AI-powered applications using Python, FastAPI, SQL, and REST APIs. Developed VitalFlow, a clinical intelligence platform that predicts 41 medical conditions using Machine Learning, Explainable AI, and multimodal voice/image capabilities, and HireMind, an offline recruitment platform that ranks 100,000+ candidate profiles using semantic matching and a custom multi-signal ranking engine. Passionate about building scalable, production-ready software that solves real-world problems.

Education

Godavari College of Engineering, Jalgaon | DBATU

Jul 2026

B.Tech in Computer Engineering

HSC — Sane Guruji Sec. & Higher Sec. School, Yawal

78%

SSC — K.Narkhede Vidyalaya & Kanishtha Mahavidyalaya, Bhusawal

75.20%

Experience

Edunet Foundation

Feb 2025 – Mar 2025

Artificial Intelligence Intern

- Conceptualized an AI healthcare diagnostic system during the internship later independently architected and extended into **VitalFlow**, a full scale multimodal clinical intelligence platform.
- Developed ML prototypes using Scikit-learn and built Python based AI pipelines (data validation, model training, API integration) forming the technical foundation for VitalFlow's diagnostic engine.

Projects

VitalFlow — AI-Powered Clinical Intelligence Platform

Python · Scikit-learn · Groq API · Gradio · SQLite

- Architected an end-to-end clinical decision support application predicting **41 medical conditions** from **132 structured symptom features** using a **Calibrated Random Forest** model.
- Engineered a **Dynamic Symptom Recommendation** engine that analyzes symptom co-occurrence patterns and continuously updates recommendations using real-time Machine Learning probability rankings.
- Improved prediction reliability by calibrating Random Forest probability estimates using **CalibratedClassifierCV**, producing more trustworthy confidence scores.
- Integrated **Groq Whisper** (speech-to-text) and **Llama 3 Vision** (image understanding) for multimodal symptom collection, and built automated PDF reports (fpdf2) with SQLite-logged prediction history.

HireMind — Offline AI Recruitment & Candidate Ranking Platform

Python · Streamlit · Sentence Transformers · Pandas · NumPy · Pytest

- Built an offline AI recruitment platform processing and ranking **100,000+ candidate profiles** via semantic skill matching and multi-signal evaluation, with no dependency on external LLM APIs.
- Designed a **5-signal ranking engine** (Role Fit, Semantic Skills, Hiring Intent, Career Growth, Stability) using Sentence Transformers (all-MiniLM-L6-v2) to generate a recruiter-ready **Top 100 shortlist** with explainable scoring.
- Engineered a data validation pipeline that filtered **18,899 suspicious profiles** and built an interactive Streamlit dashboard with score breakdowns, recruiter verdicts, and Pytest backed testing.

Full-Stack Agentic AI Platform

Python · FastAPI · LangGraph · LangChain · Streamlit

- Built a full-stack Agentic AI platform integrating **Groq**, **OpenAI**, and **Tavily** for real time AI responses.
- Designed a **ReAct-based workflow** via FastAPI REST APIs for tool augmented, multichat, voice conversations.

Technical Skills

Languages: Python, SQL, JavaScript

Backend: FastAPI, RESTful APIs, API Integration

AI & Machine Learning: Scikit-learn, LangChain, LangGraph, Sentence Transformers, Prompt Engineering, NLP, XAI, Model Calibration

Databases: PostgreSQL, SQLite

Tools: Git, GitHub, Streamlit, Gradio, Pandas, NumPy, Pytest

Cloud & AI Services: Render, Hugging Face Spaces, Groq API, OpenAI API, Tavily API